
What does self-supervision add to engineered features for stellar light curves?

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Abstract

TESS light curves encode a star’s pulsation, rotation, binarity and oscillations, among other physical properties, but the survey produces far more light curves than can be labelled by hand. Astronomers have engineered a few dozen physically motivated features from them, which together remain a strong baseline. Previous work has trained encoders on light curves and judged them by their own downstream scores, but has not asked what they add to the engineered features. We pre-train a convolutional encoder on unlabelled TESS light curves to predict each window’s latent representation from those of its neighbours. We then freeze it, add its features to the 25 engineered features, and read the combination out with one fixed linear probe on ten tasks: five variability classes, four asteroseismic targets and rotation period. Adding the learned features improves the readout on 7 of 10 tasks over the engineered features alone, where an untrained encoder does not. The combination also outperforms two networks trained end-to-end on the same labelled stars on most tasks. An encoder trained without labels therefore gives astronomers new features that complement the ones they already use rather than replace them.

1 Introduction

A star’s light curve carries its pulsation, rotation, eclipses and oscillations, and with them its age, interior and companions. Wide-field surveys now produce light curves far faster than they can be labelled. TESS [Ricker et al., 2015] returns 2-minute or 20-second [Huber et al., 2022] light curves for hundreds of thousands of pre-selected stars, and full-frame images of every star in the field at cadences that change between mission cycles (30 minutes, then 10, now 200 seconds). LSST [Ivezić et al., 2019] will add *billions* more light curves, albeit at a cadence of days. Self-supervised learning is the natural response: pre-train one encoder on the unlabelled archive, then attach a cheap readout for each task. Encoders of this shape exist for ground-based, Gaia and Kepler light curves [Donoso-Oliva et al., 2025, Zuo et al., 2025, Rui, 2026] and are evaluated by classification accuracy, sometimes against engineered features [Rui, 2026]; a label-free TESS representation [Poznanski, 2026] is evaluated by whether repeat observations of one star lie close in its embedding space. None measures what they add to engineered features.

In this domain, the baseline that a learned encoder must improve on is already strong. Decades of variable-star research have distilled light curves into compact engineered summaries: periodogram peak structure, band-limited power, spectral entropy and robust scatter statistics. The benchmark of Li et al. [2025] makes the point sharply: across seven expert-labelled variable-star classes, no time-series

foundation model they tested strictly surpassed the long-established domain feature baseline on classification. A learned encoder for this data therefore has to answer one question: do its features add predictive signal to the engineered features under the same readout?

We measure that directly. We pre-train an encoder on TESS light curves, freeze it, and add its features to the 25 engineered features under one fixed linear readout. Our contributions are:

- **The learned features add to the engineered features rather than replacing them.** Adding the learned features to the engineered ones improves a fixed linear readout beyond 2 SE on 7 of 10 tasks, on each of which an untrained encoder gives no such gain, so the gain comes from pre-training, not from merely adding 128 columns.
- **The combined pipeline, read out by a fixed linear model, outperforms the two supervised networks we train end-to-end on the same labels on most tasks.** Against a supervised Conv1D with the encoder’s architecture it is ahead on 7 of 10 tasks and indistinguishable on the other 3; against an MLP on raw flux it is ahead on 8 of 10.

2 Data and method

Data. We use 2-minute SPOC PDCSAP light curves of stars brighter than TESS magnitude $T = 10$ from Sectors 27–101. Pre-training and three of the ten probes (pulsating, eclipsing binary and transit) use a label-enriched subset of 13,470 stars, split 70/15/15 by star: every star labelled as a transit host, eclipsing binary or pulsator that has a usable window, plus a random sample of 10,000 quiet stars, listed in none of our variability catalogues. The pre-training loss never sees a label, but this pool is label-enriched, not a random draw from the survey. Five probes draw their labels from catalogues that barely overlap this subset and are scored on a second pool of 22,860 stars built the same way. The two rotation probes are scored on the 106,284 corpus stars outside the first pool, split 70/15/15, where rotators are a natural 13% (Appendix J). No test star of the first pool and no star of the rotation pool enters pre-training. About 8% of the second pool’s test stars (265 of 3,429) were in the pre-training subset; removing them leaves every gain in place (Appendix I). Each light curve is split into segments wherever the time axis breaks, at a missing cadence or a jump longer than five median cadences; segments are median-normalised by their median absolute deviation and sliced into non-overlapping windows of 256 cadences ($256 \times 120 \text{ s} \approx 8.5 \text{ h}$). Windows with a missing value are discarded, never interpolated or padded, and segments are never stitched across sector or gap boundaries. Each pre-training example is 16 consecutive windows ($\approx 5.69 \text{ d}$) starting at a random point in a segment.

Encoder. A 1D convolutional variational encoder [Kingma and Welling, 2014] maps each window to a 128-dimensional Gaussian posterior, and a GRU [Cho et al., 2014] predicts the next window’s latent from its predecessors, run both forward and backward over the sequence. The objective is

$$\begin{aligned} \mathcal{L} = & \|\hat{x} - x\|^2 + \beta \text{KL} + \lambda [\|\hat{z}_{t+1} - \bar{z}_{t+1}\|^2 + \|\hat{z}_{t-1} - \bar{z}_{t-1}\|^2] \\ & + w [\|\log P_H(\hat{x}) - \log P_H(x)\|^2 + \|\Delta \hat{x} - \Delta x\|^2], \end{aligned}$$

where every term is a mean squared error, $\bar{z}$ the stop-gradient target, P_H the Hann-tapered power spectrum, Δ the first difference, and $(\beta, \lambda, w) = (0.1, 60, 0.3)$, β warmed up linearly over ten epochs. The encoder is trained with six seeds and then frozen; no gradient from any label reaches it.

Representations and readout. A star’s learned features are μ , the mean, over every window of its first segment (16 for 95% of stars, 20 for the rest), of the per-window posterior means. The engineered baseline is a fixed 25-feature vector modelled on the T’DA variability pipeline [Audenaert et al., 2021], computed on the same first segment, 19 from its periodogram and 6 time-domain statistics; Appendix H defines each feature. Readouts are fixed in advance and never tuned on a task, with inputs standardised using training-split statistics alone: class-balanced L_2 logistic regression at $C = 1$ for the detection tasks, ridge for the two regressions with its penalty set by leave-one-out cross-validation. We compare four arms, each the same readout on a different feature set: **features**, μ , **features** $\oplus \mu$, and, as a control, the same fusion built on an *untrained* encoder; the two supervised baselines below are the fifth and sixth arms. Appendix D repeats the first three under XGBoost.

Supervised baselines. Two end-to-end baselines see exactly the same input windows, star sets and splits. **C1** reuses the encoder’s convolutional trunk unchanged, adds a 128-unit linear layer

Table 1: Ten-task scorecard. Frozen arms: a fixed logistic or ridge readout, or XGBoost on the same inputs (Appendix D). Bold: best point estimate of the three arms under each frozen readout, with no significance test (appendix tables bold only differences beyond 2 SE). n : test stars; prev.: positive-class prevalence. The two rotation rows are scored on the rotation pool at the survey’s own rotator fraction; the other detection rows are case-control.

task	metric	n (prev.)	frozen linear readout			frozen XGBoost readout			end-to-end supervised	
			features	μ	features $\oplus \mu$	features	μ	features $\oplus \mu$	Conv1D	MLP
pulsating	PR-AUC	2021 (0.107)	0.789	0.806	0.846	0.845	0.821	0.834	0.816	0.748
eclipsing binary	PR-AUC	2021 (0.097)	0.742	0.771	0.775	0.798	0.789	0.807	0.774	0.719
rotation	PR-AUC	15944 (0.130)	0.386	0.373	0.441	0.518	0.436	0.515	0.409	0.420
transit	PR-AUC	2021 (0.060)	0.190	0.144	0.218	0.217	0.223	0.228	0.134	0.109
oscillating giant	PR-AUC	3429 (0.383)	0.920	0.854	0.918	0.924	0.899	0.923	0.875	0.883
solar-like osc.	PR-AUC	3429 (0.122)	0.320	0.336	0.390	0.470	0.409	0.463	0.396	0.275
RGB vs. HeB	ROC-AUC	161 (0.702)	0.758	0.660	0.708	0.746	0.725	0.763	0.530	0.762
$\nu_{\max}$	R^2	1313	0.831	0.802	0.867	0.916	0.831	0.919	0.832	0.834
rotation period	R^2	1318	0.718	0.636	0.752	0.829	0.729	0.825	0.697	0.773
flare activity	PR-AUC	3429 (0.089)	0.474	0.453	0.524	0.562	0.485	0.565	0.533	0.444

and a scalar output, and is trained per task with a class-weighted loss on the mean of the star’s window logits, computed inside the network so that the gradient flows through the mean. **C2** replaces the convolutional trunk with a dense one and changes nothing else, so C1 versus C2 isolates the convolutional inductive bias. Both use one regularisation setting, fixed in advance, for all tasks and select on a validation split no probe has ever touched.

Tasks and metrics. We score ten tasks with one probe per physical quantity: pulsating variables [Gao et al., 2025], eclipsing binaries [Prša et al., 2022], rotation and rotation period [Boyle et al., 2026], transits [Guerrero et al., 2021], flare activity [Seli et al., 2025, Vida et al., 2021], oscillating red giants and their $\nu_{\max}$ [Hon et al., 2021], solar-like oscillators [Hatt et al., 2023], and RGB versus helium-burning discrimination [Sreenivas et al., 2026]. Here $\nu_{\max}$ is the frequency of maximum oscillation power. Detection tasks are scored by the area under the precision–recall curve (PR-AUC), whose floor is the positive-class prevalence rather than 0.5; prevalence is therefore quoted with every detection score. RGB versus helium-burning separates two red-giant populations rather than detecting a rare class, so it is scored by ROC-AUC; the two regressions are scored by R^2 .

Uncertainties. All error bars are $2\times$ the standard error of the mean over seeds. For the μ arms the seeds are independent pre-training runs (six); for C1 and C2 they are initialisation and shuffling seeds (six); the engineered arm is deterministic. The μ and supervised seeds are unrelated, so differences between those arms are unpaired, with bands combined as $\sqrt{\text{SE}_a^2 + \text{SE}_b^2}$. A difference counts only beyond its 2 SE band. These bands measure re-training spread only; a paired bootstrap over the test stars (1,000 resamples, seeds averaged; Appendix G) checks that no verdict hinges on the test sample.

3 Results

The learned features add to the engineered features. Adding μ to the 25 features improves the fixed readout (Table 1, Figure 1a) beyond 2 SE on 7 of 10 tasks, led by solar-like oscillators $+0.071$, pulsating $+0.057$ and flare activity $+0.051$, and reaching $+0.034$ (R^2) on rotation period. Rotation clears the bar too ($+0.055 \pm 0.006$) but does not count: an untrained encoder gains about half as much there ($+0.027 \pm 0.003$ over six random initialisations), and a gain counts only if the untrained control’s gain stays within 2 SE of zero, a rule fixed before the rotation tasks were scored (Appendix J). The two negative tasks are those where 128 extra columns cost more than they add: the 25 features already cover the task (oscillating giants, -0.003 on a baseline of 0.920) or the training set is too small to fit 128 more weights (RGB versus helium-burning, -0.050 on 755 training stars). On both, an untrained encoder loses about as much, within the error band, so redundant columns, not misleading content in μ , cause the loss (Appendix C). On the other nine tasks the untrained control never exceeds $+0.016$ and is negative on six (Table 2); outside the two rotation tasks it is one random encoder, with no error band. So the gains that count come from pre-training, not from 128 extra columns. μ on its own does not replace the features (ahead on 3 of 10 alone), as Li et al. [2025] find for foundation models; we therefore report fusion rather than substitution [Makinen et al., 2024].

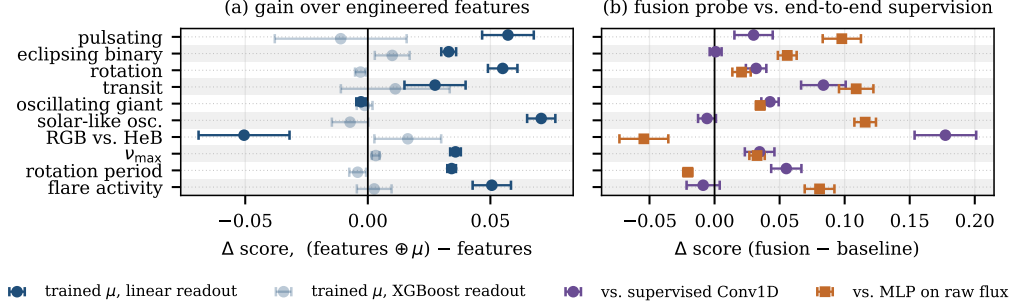


Figure 1: The two claims. **(a)** What the learned features add to the engineered features, under the fixed linear readout of Table 1 (solid) and under an XGBoost readout on the same inputs (faded; Appendix D). **(b)** The fusion probe against the two supervised baselines. Bars are 2 SE.

The fusion probe outperforms the two supervised networks on most tasks. Against C1, which shares the convolutional trunk, labels and input windows but is trained end-to-end, the fusion probe is ahead on 7 of 10 tasks and indistinguishable on the other 3 (Figure 1b). Against C2, an MLP on raw flux, it is ahead on 8 of 10 and behind on two (RGB versus helium-burning and rotation period). C1 is not weak: it beats the engineered arm on 5 of 10. A paired star-bootstrap (Appendix G) excludes zero on 6 of the 7 fusion gains that count, 6 of the 7 leads over C1 and all 8 over C2. This holds for this architecture on 755 to 74,398 labelled stars per task, not for supervision in general.

The GRU prediction term raises the gain on every task. Retraining without it, all else unchanged, lowers the fusion gain on all 10 tasks, by more than 2 SE on 6 when compared seed by seed (Table 2).

4 Discussion

Pick the encoder by a held-out probe, not by its loss. We trained 29 variants of the pre-training recipe that differ in their loss terms, and scored each on the three detection tasks of the first star pool, the tasks we developed the recipe on (Appendix E). On none of the three did the variant with the lowest validation reconstruction loss score best, and choosing it would have cost 0.011–0.031 PR-AUC. An encoder therefore has to be chosen by a held-out probe, not by its own loss.

Limitations and next steps. *Readout.* The four linear arms share one small linear model, which cannot rescue weak features, so the comparison is between feature sets. Our claim is confined to these 25 features and a linear readout: under XGBoost the gain exceeds 2 SE on 3 of 10 tasks and is negative on two, and it shrinks most where XGBoost improves most on the features alone (Spearman 0.87; Appendix D). *Data.* Every arm, the supervised baselines included, sees one 5.7-day segment of a bright 2-minute star; fainter full-frame-image stars and multi-sector light curves come next. *Labels.* Our negatives are stars absent from the catalogues, not stars verified to lack the signal, so catalogue incompleteness adds positive-unlabelled label noise, and detection scores may be conservative. The flare label marks stars that flared anywhere in their TESS data, so the task is recognising a flaring star, not finding a flare in the encoded segment. *Label scarcity.* With the readout trained on a few percent of the labels, the advantage narrows on 7 of 10 tasks, so how much pre-training helps when labels are scarce remains open. *Reconstruction.* The decoder plants a narrow spike near the centre of each window; every loss change we tried that removed it also hurt some probe scores (Appendix F). *Discovery.* Every task here has a catalogue to score against; whether μ also groups stars into kinds of variability no catalogue lists yet is untested and needs a different evaluation.

Reproducibility and disclosure. §2 and Appendices A and H give the full recipe; code, configuration files and every score table are at <https://github.com/BrightonLiu-zZ/stellar-ssl/tree/arxiv-v1>. Generative AI assisted with analysis code and prose editing.

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A Pre-training and readout details

Architecture. The encoder has four convolutional stages (widths 32/64/128/256, kernel 5, each halving length) and a 128-dimensional latent; the predictor is a one-layer GRU of width 256.

What one pre-training example is. The packed pre-training corpus is the 70% training split of the first pool: 9,428 stars, 35,854 segments and 582,352 windows of 256 cadences. The encoder is monitored on the pool’s 15% validation split (2,021 stars, 121,328 windows), which no probe ever sees. One example is one segment. Each epoch it yields a fresh run of 16 consecutive windows beginning at a uniformly random offset inside that segment, so a long segment shows a different 5.69-day stretch every epoch; a segment with fewer than 16 windows never entered the pack. Validation uses offset 0 and every segment, so the monitored quantity does not move with the sampling. A batch is 256 sequences, giving 140 optimiser steps per epoch.

The first-difference term. In the objective of §2, $\Delta x_t = x_{t+1} - x_t$ is the first difference along a window, and $\|\Delta \hat{x} - \Delta x\|^2$ is the mean squared error between the reconstruction’s cadence-to-cadence changes and the input’s. It is there because the point-wise term $\|\hat{x} - x\|^2$ is already small for a smooth curve drawn through a noisy one: the fastest components of a light curve carry little amplitude, so a decoder that low-passes its input pays almost nothing for discarding them. Differencing is a high-pass operation, so the same error re-enters the objective weighted towards high frequency and the smoothed reconstruction is penalised. P_H plays the same role in the frequency domain; the two share the weight w and are summed with equal weight inside it.

Optimisation. Adam at learning rate 3×10^{-4} , cosine-annealed to 3×10^{-6} over the 100-epoch cap, gradient-norm clipping at 1.0, mixed precision. β rises linearly from 0 to its target 0.1 over the first ten epochs and stays there; the shipped recipe uses no free-bits floor. Training stops early once ten consecutive epochs improve no tracked checkpoint, a counter that is armed only after the warm-up.

Which checkpoint is frozen. Only post-warm-up epochs are eligible. During the warm-up β and the KL term are still moving, and the scheduled- β total is minimised at $\beta = 0$, that is by an untrained model, so an unrestricted argmin can select the warm-up transient rather than a fit. Among the eligible epochs we freeze the one minimising recon + $w \cdot$ aux on the validation split. The prediction term is deliberately absent from that metric: it is the quantity a dynamics ablation changes, so including it would select each arm of that comparison by a different rule.

Readouts. Logistic regression is scikit-learn’s LogisticRegression with L_2 penalty, $C = 1$, class_weight="balanced" and at most 2,000 iterations. Ridge is RidgeCV over ten log-spaced penalties from 10^{-2} to 10^3 , chosen by leave-one-out cross-validation. Every input column is z-scored with the training split’s own mean and standard deviation, and the same two numbers are applied unchanged to validation and test. For the fusion arm the 25 features and the 128 latent means are column-concatenated first and the 153-column matrix is standardised as a whole, so no arm gets a different scaling of the columns it shares with another.

The supervised baselines. C1 and C2 are trained with AdamW at learning rate 3×10^{-4} and weight decay 10^{-4} , dropout 0.2 before the task head only (the trunk is unregularised, as the encoder’s is), gradient clipping at 1.0, mixed precision, batches of 64 stars ($\approx 1,040$ windows, since the loss is per star), at most 60 epochs and patience 10. The positive class enters the loss with weight $n_{\text{neg}}/n_{\text{pos}}$ computed on the training split. One regularisation setting is fixed in advance for all ten tasks and no per-task tuning is done. Each run is selected on the best validation score in that task’s own headline metric rather than on validation loss: at the prevalences here the validation-loss minimum and the validation-metric maximum routinely fall many epochs apart, and selecting on the loss would depress the baseline we are trying to beat.

B The two star pools, and what a quiet star is

Each task is scored on the pool its catalogue populates, and every arm on a task sees the identical star list, so the two definitions below fix the population but never a comparison.

The first pool (13,470 stars). Every star in the corpus with at least one usable window that carries a transit, eclipsing-binary or pulsating label: 3,470 stars, assigned to a single split stratum rarest class first (813 transit, 1,226 eclipsing binary, 1,431 pulsating). To them are added 10,000 *quiet* stars drawn uniformly at random, without replacement, from the corpus stars that appear in none of our five v1 catalogues – transit, eclipsing binary, pulsating, rotation and flare. The whole pool is split 70/15/15 by star within each stratum, and the encoder is pre-trained and monitored on this pool’s training and validation splits.

Rotation and flare appear here only as exclusions on the quiet draw, so the rotation tasks are scored on a pool of their own (Appendix J).

The second pool (22,860 stars). The four external catalogues behind the asteroseismic and rotation-period tasks are joined to the corpus into one table of 11,007 stars, each row a star that at least one of them covers. Those 11,007, plus the 1,853 flare-labelled corpus stars not already among them, are the pool’s members (12,860). To them are added a fresh draw of 10,000 quiet stars: corpus stars that carry no transit, eclipsing-binary, pulsating or rotation label, have no flare label, and are absent from the external table. The split is again 70/15/15 by star, with the stratum taken rarest first over RGB-versus-helium-burning, Kounkel rotation period, flare, solar-like oscillator, oscillating giant and quiet.

The two screens are not symmetric. The first pool’s quiet draw is screened against the five v1 catalogues only; it never consults the four external ones, so some of its quiet stars are oscillating red giants or solar-like oscillators. The second pool’s quiet draw is screened against both sets. Nothing printed here compares a star in one pool with a star in the other, so the asymmetry costs no verdict; what it does mean is that a negative in either pool is a star *absent from the relevant catalogues*, not a star examined and found to lack the signal. The external table is itself a union of four detection lists, so a star it covers with a zero for oscillating giants is a star Hatt, Sreenivas or Kounkel listed and Hon et al. [2021] did not, which is the same kind of negative. This is the label limitation §4 states, given exactly.

C The two tasks where fusion does not help

task	features	fusion Δ	2 SE	untrained Δ	n_{test}
oscillating giants	0.920	−0.003	0.002	−0.003	3,429
RGB vs. He-burning	0.758	−0.050	0.019	−0.038	161

Oscillating giants: the 25 features already cover the task. The labels of Hon et al. [2021] come from a convolutional classifier reading a 128×128 image of each star’s power spectrum, which fires on a visible oscillation power excess at $6 \lesssim \nu_{\text{max}} \lesssim 250 \mu\text{Hz}$. Our engineered vector encodes close to that same quantity: eight log-spaced band powers spanning 0.3–360 c/d, plus the spectral centroid and entropy, locate the excess and describe its concentration. The features accordingly reach 0.920,

and the trained and untrained fusion arms then lose the same 0.003: 128 collinear columns diluting a linear fit, with no trained-versus-untrained difference left to attribute to μ .

RGB versus helium-burning: no arm resolves the discriminant. Sreenivas et al. [2026] assign the evolutionary state with a convolutional classifier reading the background-corrected power spectrum *folded at* $\Delta\nu$; the physics underneath is the dipole mixed-mode period spacing $\Delta\Pi_1$ [Bedding et al., 2011]. Neither is accessible at our baseline. The encoder sees one 256-cadence window at a time, so the finest frequency structure available to μ is $\Delta f = 32.6 \mu\text{Hz}$, while $\Delta\nu$ for these stars is $\approx 1.5\text{--}18 \mu\text{Hz}$; the comb itself is unresolved, let alone a fold of it. The engineered features, computed on the whole 5.69 d segment, reach $2.0 \mu\text{Hz}$, still coarser than the mixed-mode spacing $\Delta\Pi_1\nu^2 \approx 0.2\text{--}0.6 \mu\text{Hz}$ near $\nu_{\text{max}} = 50 \mu\text{Hz}$. No arm on this row measures the quantity the label encodes; the 0.758 the features reach rides on correlates of evolutionary state rather than on its discriminant. The row behaves accordingly: it is the only one where C2 (0.762) is ahead of the fusion probe while C1 reaches only 0.530 on 755 training stars, and it carries the smallest test set in the study.

Why 128 redundant columns cost a linear fit. A linear readout spends its training stars estimating one coefficient per column, and each estimate carries its own error into the prediction whether or not the column carries signal. On the RGB-versus-helium-burning row the features arm fits 25 coefficients on 755 training stars and the fusion arm 153, so the stars-per-coefficient ratio falls from about 30 to about 5 while, by the resolution argument above, the added columns supply nothing the label depends on. The variance a column adds scales with how few stars are available to pin it down; the signal it adds does not. A column is therefore worth its place only when what it adds exceeds that cost, and on this row and on oscillating giants it does not. The logistic readout’s L_2 penalty is fixed at $C = 1$ for every task rather than tuned per task, so it damps this effect but does not remove it. The prediction that follows is the one the table above shows: an untrained encoder’s 128 columns should cost about the same as a trained one’s, since the argument never mentions what the columns contain, and they do (−0.038 against −0.050, and −0.003 against −0.003).

D The gain under a nonlinear readout

Table 2: What 128 added columns change, relative to the 25 features alone, under each frozen readout (mean over six encoder seeds; absolute scores are in Table 1). $+\mu$: our encoder. No prediction term: the same recipe trained without the GRU prediction term. Retained: the share of the full μ gain that survives that removal, printed only where the full gain is a gain, and negative where the gain changes sign rather than shrinking. Untrained: a randomly initialised encoder, which is a single encoder and carries no band. Bold: beyond 2 SE — unlike Table 1, where bold marks the best point estimate and implies no test. The last row counts tasks won, tied and lost at that bar.

task	metric	linear readout				XGBoost readout	
		$+\mu$	$+\mu$, no prediction term	retained	$+\text{untrained } \mu$	$+\mu$	$+\text{untrained } \mu$
pulsating	PR-AUC	+0.057	+0.051	0.89	+0.016	−0.011	−0.003
eclipsing binary	PR-AUC	+0.033	+0.012	0.38	−0.010	+0.010	−0.003
rotation	PR-AUC	+0.055	+0.036	0.65	+0.029	−0.003	−0.010
transit	PR-AUC	+0.027	+0.018	0.66	+0.004	+0.011	−0.027
oscillating giant	PR-AUC	−0.003	−0.006	−	−0.003	−0.001	−0.008
solar-like osc.	PR-AUC	+0.071	+0.024	0.33	−0.007	−0.007	−0.026
RGB vs. HeB	ROC-AUC	−0.050	−0.052	−	−0.038	+0.016	−0.031
ν_{max}	R^2	+0.036	+0.010	0.29	−0.001	+0.003	−0.005
rotation period	R^2	+0.034	+0.014	0.40	−0.003	−0.004	−0.013
flare activity	PR-AUC	+0.051	+0.033	0.65	+0.006	+0.003	+0.006
win / tie / loss		8 / 0 / 2				3 / 5 / 2	

The retained column is the per-task form of the body’s ablation count. On the eight tasks where fusion gains, removing the GRU prediction term leaves between 0.29 and 0.89 of that gain, below half on four of them; on none does the gain change sign. We give the eight fractions rather than one summary number because the ten gains are measured in PR-AUC, ROC-AUC and R^2 and cannot be added together.

The XGBoost readout takes the same inputs as the linear one. For each task and seed it is selected over a grid of maximum depth $\{3, 5\}$ and learning rate $\{0.05, 0.1\}$, with up to 1,000 rounds and early stopping after 50 rounds without improvement on a held-out 25% of the training stars; the test stars play no part in the selection. Under XGBoost the features-only arm also varies with the seed, through the validation split, so the gain is differenced seed by seed.

Under XGBoost the μ gain (features $\oplus \mu$ minus features alone) is smaller than under the linear readout on most tasks. XGBoost also scores higher than the linear readout on the 25 features alone on nine of ten tasks, and the two changes go together. On solar-like oscillators the features-only score rises from 0.320 to 0.470 and the μ gain falls from +0.071 to -0.007 . On RGB versus helium-burning, where XGBoost does not raise the features-only score, the μ gain is larger under XGBoost. Across the ten tasks, the rise in the features-only score and the fall in the μ gain have a Spearman rank correlation of 0.87. This is consistent with part of the information μ supplies to a linear readout being already present in the 25 features, in a form only a nonlinear readout can use.

Table 2 also gives the same comparison with an untrained encoder in place of the trained one. Under XGBoost its 128 columns lower the score on nine of ten tasks and the trained μ scores above it on eight, so the gains that survive come from pre-training, not from adding columns. XGBoost on the 25 features alone also scores above the linear fusion probe on 8 of 10 tasks (Table 1). The body’s claim is about what μ adds under a fixed linear readout, not that the linear fusion probe is the strongest classifier available on these features.

E Validation reconstruction against downstream score

The 29 recipes behind §4 come from two rounds of pre-training experiments, one per figure. Figure 2 is the earlier round, ten recipes that differ only in the GRU prediction term; Figure 3 is 19 variants of our final recipe. Every recipe was scored on the three detection tasks of the first star pool, the tasks the recipes were developed on, so those are the tasks shown. The other seven probes were not run for every recipe, and the rotation tasks are scored on a pool these recipes were never encoded on. Pooled over all 29 recipes, the one with the lowest validation reconstruction loss is the same on all three tasks (the diamond in Figure 3, which has no spectral auxiliary term); it ranks 10th, 27th and 19th of 29 on eclipsing binaries, pulsators and transits, and trusting it costs 0.011–0.031 PR-AUC against the best recipe. Each figure below repeats the comparison within its own round.

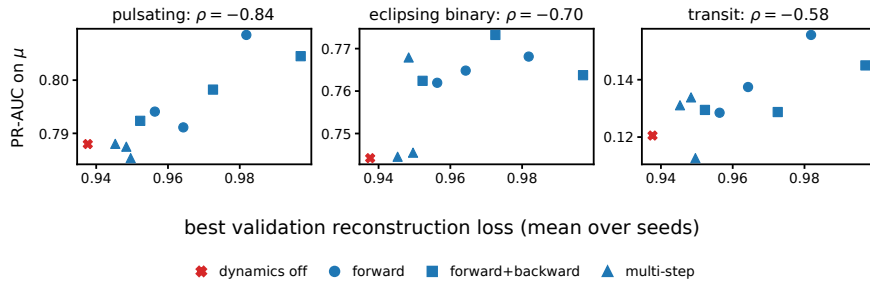


Figure 2: Validation reconstruction loss against downstream probe score in the earlier round, one point per pre-training recipe and one panel per task. The ten recipes share corpus, architecture and loss terms and differ only in the GRU prediction term (off, forward, forward+backward, multi-step, the last three at three weights), four seeds each. x : best post-warm-up validation reconstruction loss, averaged over the recipe’s seeds; y : PR-AUC of the fixed linear probe on μ ; ρ : Spearman’s rank correlation between negated loss and score. The loss-optimal recipe is the dynamics-free one (red) on all three tasks, and it is the worst or next-to-worst probe on each.

In Figure 2 the loss-optimal recipe is the dynamics-free one on all three tasks, and it is the worst or next-to-worst probe on each, so selecting on the loss costs 0.021–0.035 PR-AUC against the probe-optimal recipe. The trend is not carried by that one point: with it removed, ρ still runs from -0.47 to -0.87 . Those recipes predate our encoder – they carry an earlier spectral auxiliary term – so Figure 3 repeats the estimator on 19 variants of the final recipe: 16 vary only the form, weight

or floor of the spectral auxiliary term, two also put a floor under the KL term, and one removes the GRU prediction term; six seeds each (two more variants in which one seed collapsed are excluded; we never drop a seed).

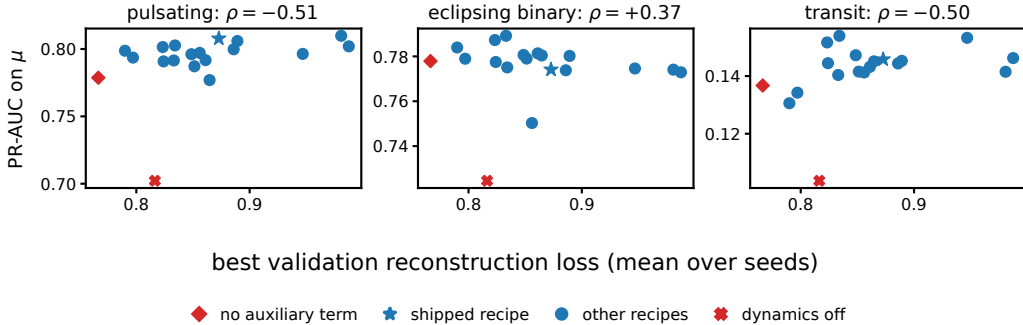


Figure 3: The same estimator as Figure 2, on the 19 variants of our final recipe. The diamond has no auxiliary term, the cross is the one variant without the GRU prediction term, the star is the recipe we use. x : best post-warm-up validation reconstruction loss; y : PR-AUC of the fixed linear probe on μ ; ρ : Spearman’s rank correlation between negated loss and score.

There the anti-correlation does not persist as such: ρ is -0.51 on pulsating and -0.50 on transit but $+0.37$ on eclipsing binaries. What persists is that the loss is not a selector: its optimum is again the recipe with the weakest competing term (the diamond, no auxiliary term at all), which is never the probe optimum, and trusting it costs 0.011–0.031 PR-AUC. So Figure 2 shows an anti-correlation on every task; this one shows it surviving on two tasks and reversing on one; in neither does the loss choose. Both are statements across recipes only: within one recipe, across its seeds, no label-free curve metric we tried (reconstruction, reconstruction plus auxiliary, active latent units) ranks the seeds, every such correlation sitting within ± 0.3 of zero.

F The within-window artifact, and the two gates that governed it

Why the objective carries a tapered power spectrum. A point-wise reconstruction error prices a light curve by amplitude, and in these data amplitude sits mostly in the slow component, so a decoder can score well on it while discarding the periodic structure the tasks depend on. The two auxiliary terms put the pressure back where the point-wise term does not reach: the first difference in time, and the log power spectrum across frequency. The spectrum is taken on a Hann-tapered window rather than on the raw one, for the reason the rest of this appendix gives.

What the decoder found instead. Every term in the objective is computed inside a single 256-cadence window; nothing spans a boundary. A narrow impulse raises log-power at every frequency at once, so a decoder can lower the log-spectrum error by planting one spike rather than by reproducing the star’s spectrum. Where the cheapest spike sits is decided by the taper, a fixed function of position within the window: with a rectangular window it is the window edge, and under a Hann taper it is the window centre, where the taper weight is largest. Figure 4 shows both, in flux and in the position-resolved error. Laid end to end, the spikes form one train with a period of exactly one window, which is a property of the display and not of the objective.

Measured as the per-position reconstruction error divided by its own interior median (positions 16–240), maximised over position within each seed and then averaged over six seeds, the rectangular recipe reaches $31.7\times$ at the window edge and the tapered recipe $11.8\times$ near the centre. The taper moved the defect and shrank it; it did not remove it. An earlier claim that it had was based on a measurement taken only at the edge, and is withdrawn.

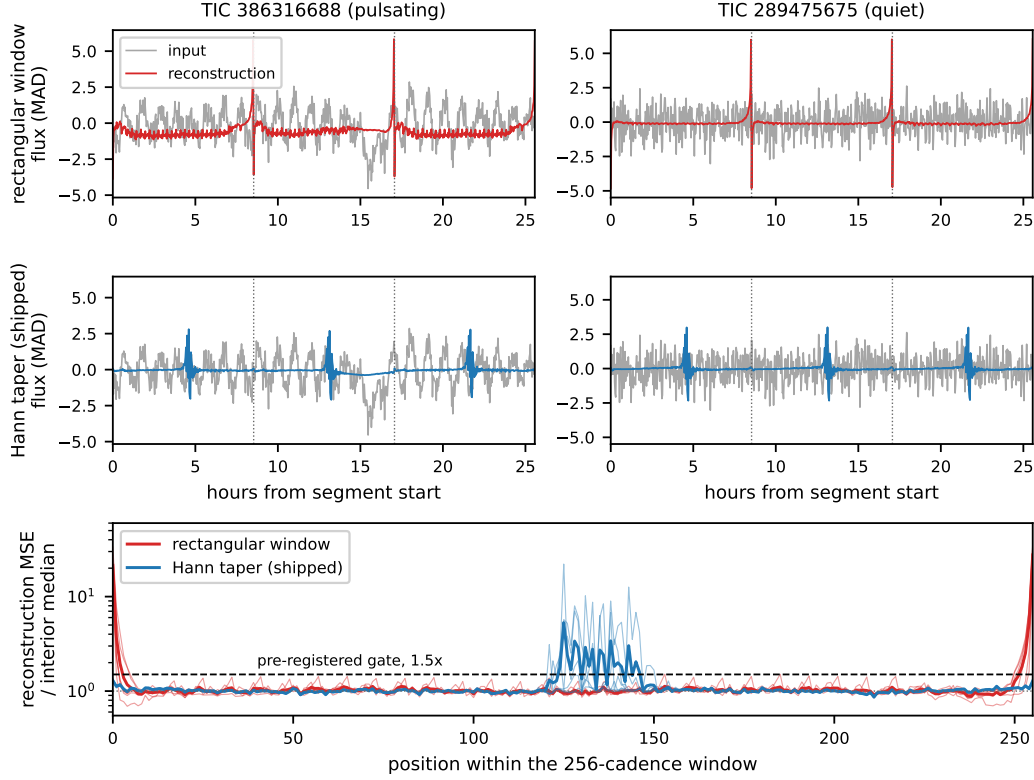


Figure 4: The artifact, in two stars and in the average. **Top two rows:** three consecutive 256-cadence windows of one pulsator and one quiet star, input flux in grey against each recipe’s reconstruction; dotted lines mark the window boundaries, and both rows of a column share one y -scale. The two recipes differ only in the taper used for the auxiliary spectral term. Neither decoder reproduces the cadence-scale scatter, which a 128-dimensional bottleneck cannot carry; what differs is where each one puts a spike it does not need. **Bottom:** reconstruction error by position within the window, divided by that seed’s median over positions 16–240. Thin lines are the six pre-training seeds, thick lines their mean. The dashed line is the pre-registered artifact gate. The seed mean understates the tapered recipe’s peak because its position moves between seeds (125–143), which is why the gate is read as a maximum within each seed.

The two pre-registered acceptance criteria. Both were registered on 2026-08-15, before any of the interventions below were written.

Artifact gate. The within-window impulse ratio – the quantity plotted in the lower panel of Figure 4 – taken as the maximum over *all* positions rather than at the edge alone, at most $1.5\times$ at full auxiliary weight, over six seeds, *and* a visual sign-off on a fresh random sample of reconstructed light curves. The number never passes on its own: reading an edge-only number is how the withdrawn claim above was made in the first place.

No-regression gate. No task among eclipsing binaries, pulsators, rotation and transits worse than the incumbent recipe by more than 2 SE, at both readouts, paired over six seeds. It is the guard against a clean but useless encoder: a recipe that abolishes the impulse and loses the probe is not a fix.

What we tried, and why the tapered recipe is the one we ship. We varied the auxiliary term along four axes: its *mechanism* (remove it entirely; average the spectrum over a family of Slepian tapers so that no single position is cheapest; add a penalty on the excess kurtosis of the residual; both together; put a floor under the spectral sub-term), its *amount* (six weights of the best mechanism, from 0.0125 to 0.20), the *level* of that floor (rebuilt as a two-sided hinge, which unlike a clamp keeps a restoring gradient, at three measured values), and finally a *floor on the KL term*, which was the

remaining untried lever. No cell cleared both gates. The ones that removed the impulse lost the probe; the ones that kept the probe kept a measurable impulse. The nearest miss reached 1.55 ± 0.05 , which is 2.4 SE above the gate; its repaired variant reached 1.47 over six seeds and 1.55 over twelve, and regressed on pulsators at both readouts.

Because no candidate passed, the rule fixed before the ladder ran leaves the incumbent in place. The encoder this paper uses is therefore the Hann-tapered recipe by a pre-registered fallback, not because it won a comparison. Two things make that defensible rather than merely procedural. The artifact was separately measured not to cost the downstream probe – ablating it moved $\nu_{\max}$ by $+0.0001 \pm 0.0037$ – and closing the channel, which was also a test of whether the validation loss becomes a usable selector once it can no longer be lowered by the trick, did not make it one (Appendix E).

G Paired bootstrap over test stars

The 2 SE bands in the body come from re-training and say nothing about whether the particular test stars favoured one arm. For every printed contrast we therefore resample each task’s test set with replacement 1,000 times, rescore every arm on the identical resampled list from its saved per-star predictions (no model is retrained), average each arm over its seeds, and take the difference. Table 3 gives the 95% percentile interval of that difference and the share of resamples in which the fusion arm is ahead; intervals that exclude zero are bold. The check tightens no verdict and reverses none: every row the body calls ahead has the fusion arm ahead in at least 85% of resamples. It does loosen five of the 40 verdicts, each a resolved 2 SE verdict whose interval touches zero: the gain on transit and the loss on oscillating giants over the features, the lead over C1 on pulsating, the loss to C2 on RGB versus helium-burning (161 test stars), and the dynamics contrast on flare activity.

Table 3: Paired star-bootstrap of the four contrasts the body counts. Each entry is the 95% interval of (left arm – right arm) over 1,000 resamples of the test stars, seeds averaged, and the share of resamples with the difference above zero ($f_{>0}$; 1.00 means the left arm is ahead in every resample, 0.00 that it is behind in every resample). Bold: interval excludes zero.

task	fusion – features		fusion – C1		fusion – C2		fusion – dynamics-off fusion	
	95% CI	$f_{>0}$	95% CI	$f_{>0}$	95% CI	$f_{>0}$	95% CI	$f_{>0}$
pulsating	[+0.012 , +0.105]	0.99	[−0.008, +0.073]	0.94	[+0.045 , +0.150]	1.00	[−0.023, +0.036]	0.69
eclipsing binary	[+0.005 , +0.062]	0.99	[−0.024, +0.025]	0.52	[+0.022 , +0.094]	1.00	[+0.004 , +0.038]	0.99
rotation	[+0.043 , +0.066]	1.00	[+0.017 , +0.048]	1.00	[+0.005 , +0.039]	0.99	[+0.014 , +0.025]	1.00
transit	[−0.025, +0.065]	0.85	[+0.041 , +0.124]	1.00	[+0.046 , +0.174]	1.00	[−0.016, +0.035]	0.78
oscillating giants	[−0.010, +0.005]	0.21	[+0.028 , +0.057]	1.00	[+0.021 , +0.051]	1.00	[−0.002, +0.008]	0.87
solar-like osc.	[+0.044 , +0.097]	1.00	[−0.037, +0.022]	0.34	[+0.079 , +0.151]	1.00	[+0.025 , +0.067]	1.00
RGB vs. HeB	[−0.087, −0.012]	0.00	[+0.099 , +0.247]	1.00	[−0.126, +0.020]	0.08	[−0.037, +0.039]	0.52
$\nu_{\max}$	[+0.028 , +0.043]	1.00	[+0.024 , +0.045]	1.00	[+0.018 , +0.047]	1.00	[+0.020 , +0.031]	1.00
rotation period	[+0.023 , +0.046]	1.00	[+0.039 , +0.073]	1.00	[−0.038, −0.004]	0.01	[+0.014 , +0.027]	1.00
flare	[+0.019 , +0.078]	1.00	[−0.047, +0.029]	0.32	[+0.027 , +0.128]	1.00	[−0.000, +0.035]	0.97

H The 25 engineered features

All 25 features are computed from a star’s first segment: its windows concatenated into one flux series x of 4,096 or 5,120 cadences, already median-normalised by its median absolute deviation. The periodogram is the power spectrum of the mean-subtracted series, $P(f) = |\text{FFT}(x - \bar{x})|^2$ with the zero-frequency bin dropped, and $p(f) = P(f) / \sum_f P(f)$ its normalisation to unit sum. Frequencies are in cycles per day. f_1, f_2, f_3 are the three strongest local maxima of P , strongest first.

Table 4: The engineered feature vector, modelled on the T’DA variability pipeline [Audenaert et al., 2021].

#	feature	definition
1–3	$\log_{10} f_1, \log_{10} f_2, \log_{10} f_3$	frequencies of the three strongest periodogram peaks
4–6	p_1, p_2, p_3	normalised power at each peak, $p_k = P(f_k) / \sum_f P(f)$
7–8	$p_2/p_1, p_3/p_1$	peak-power ratios
9	harmonic ratio	$P(2f_1)/P(f_1)$ at the bin nearest $2f_1$; 0 if $2f_1$ exceeds the Nyquist frequency
10–17	band powers	$\log_{10} \sum_{f \in B_j} p(f)$ for eight log-spaced bands B_j spanning 0.3–360 c/d
18	spectral entropy	$-\sum_f p(f) \ln p(f) / \ln N$, N the number of frequency bins
19	spectral centroid	$\log_{10} \sum_f f p(f)$
20–21	skew, kurtosis	sample skewness and excess kurtosis of x
22	point-to-point scatter ratio	$\text{std}(\Delta x) / \text{std}(x)$, Δ the first difference
23	depth	95th minus 5th percentile of x
24	MAD	median absolute deviation of x
25	IQR	inter-quartile range of x

What “modelled on T’DA” means. Audenaert et al. [2021] classify Kepler Q9 light curves into eight variability classes with an ensemble of four supervised learners and a random-forest metaclassifier. We take none of that: not the classifiers, not the classes, not the training set. What we borrow is the shape of the feature vector its tree-based members read, and the borrowing is partial in ways a reader should be able to check.

Kept. A short list of periodogram peaks with their relative strengths, a harmonic measure of the dominant peak, an entropy measure of the signal, and robust time-domain scatter statistics – skewness, kurtosis and the median absolute deviation, all three of which appear in their Table 1. Our point-to-point scatter ratio $\text{std}(\Delta x) / \text{std}(x)$ is, up to a square root, their variability index η_e , the ratio of the mean squared successive difference to the variance.

Changed. They extract up to six frequencies with up to ten harmonics each by iterative prewhitening – fit a frequency and its harmonics, subtract them, repeat until a significance criterion stops the loop. We take the three strongest local maxima of one segment’s periodogram, with no prewhitening and no significance test, and keep one harmonic ratio $P(2f_1)/P(f_1)$ instead of a harmonic series. Their amplitudes and phases become, here, powers normalised by the total and ratios between them: the flux is median-absolute-deviation normalised per segment, so an absolute amplitude is not a quantity we have, and we drop phases and phase differences entirely. Their FliPer, the mean power above each of four frequency cut-offs, becomes our eight log-spaced band powers spanning 0.3–360 c/d plus a spectral centroid. Their entropy features are computed on the time series (differential entropy, multiscale sample entropy); ours is the Shannon entropy of the normalised power spectrum. Because a packed segment is uniformly sampled and gap-free, its Lomb–Scargle periodogram is the FFT periodogram, so we compute the cheaper one.

Dropped. Everything that needs a phase fold at a reliably determined period – the self-organising-map location, the folded point-to-point statistics and the folded range – because 5.69 days covers few cycles for the longer-period classes in our menagerie. Also dropped: the coherency parameter, zero crossings, the range of the cumulative sum, the Shapiro–Wilk statistic, the variance ratio, the significant-harmonic count and the flux ratio, and the power-spectrum image their deep-learning member reads, which is not a feature vector at all.

The result is 25 numbers, 19 from the periodogram and 6 from the time domain, and it is a baseline rather than a reproduction: it is deliberately the cheap, linear-readable part of a pipeline whose published accuracy came from an ensemble we do not run.

I Evaluation restricted to stars never seen in pre-training

The encoder is fitted on the training stars of the first pool and monitored on its validation stars, so the test stars of the three tasks scored on that pool are unseen by construction, and the rotation pool excludes every pre-training star. The second pool shares some stars with it. Table 5 removes every test

star that was in the pre-training training or validation split and rescores the saved per-star predictions; no model is refitted. The gain exceeds 2 SE on the same tasks as before; the largest change is flare activity, +0.051 to +0.041, and the small loss on oscillating giants (−0.003) becomes a tie (−0.001).

Table 5: The fusion gain (features $\oplus \mu$ minus features, linear readout) on all test stars and on the test stars never seen in pre-training. Bold: beyond 2 SE over six encoder seeds.

task	metric	test stars	unseen test stars	gain, all	gain, unseen
pulsating	PR-AUC	2021	2021	+0.057	+0.057
eclipsing binary	PR-AUC	2021	2021	+0.033	+0.033
rotation	PR-AUC	15944	15944	+0.055	+0.055
transit	PR-AUC	2021	2021	+0.027	+0.027
oscillating giant	PR-AUC	3429	3164	−0.003	−0.001
solar-like osc.	PR-AUC	3429	3164	+0.071	+0.069
RGB vs. HeB	ROC-AUC	161	149	−0.050	−0.048
$\nu_{\max}$	R^2	1313	1218	+0.036	+0.034
rotation period	R^2	1318	1318	+0.034	+0.034
flare activity	PR-AUC	3429	3164	+0.051	+0.041

J The rotation pool

Construction. Every corpus star with at least one usable window that is not in the first pool: 106,284 stars, of which 13,789 carry a rotation period in the TESS All-Sky Rotation Survey [Boyle et al., 2026] and 92,495 do not. By construction the pool contains no transit host, eclipsing binary or pulsator (the first pool took every one), no star that entered pre-training, and no sampling step: rotators are 13.0% of it, against 12.5% of the corpus, so both rotation tasks are scored at the survey’s own prevalence rather than the case-control prevalence of the other pools. It is split 70/15/15 by star, stratified on rotators with period at most five days (8,786), other rotators (5,003) and non-rotators, seed 0. The rotation-period task keeps the rotators with period at most five days. A non-rotator here is a star the survey does not list, not a star examined and found not to rotate. No encoder is run again: the per-star encoder outputs and the 25 engineered features for these stars had been cached for an earlier survey-prevalence rescore, and the readouts are the body’s, fitted on this pool’s training split. The design, the arms, the readouts and the rule for reading the result were fixed in writing before any score on this pool was computed.

Two properties still differ between the classes besides rotation, both astrophysical rather than sampling artefacts: 10.4% of the rotators and 0.5% of the non-rotators carry a flare label, and 0.3% of the rotators against 9.4% of the non-rotators are oscillating giants or solar-like oscillators. The headline rows keep every star, because a survey contains them; two sensitivity rows refit the readout with flare stars removed from both classes, and with the positives restricted to stars in no non-rotation catalogue and the negatives to stars in no catalogue at all.

The counting rule. A gain on these two tasks counts only if (i) it exceeds 2 SE, (ii) the untrained control’s own gain stays within 2 SE of zero, and (iii) the trained-minus-untrained difference, paired seed s with initialisation s , exceeds 2 SE. Clause (ii) needs the untrained encoder’s spread, so six random initialisations were scored here; Table 1 prints the first of them, the same initialisation as every other task’s untrained arm.

Result. Table 6 gives both tasks with the two sensitivity rows. On rotation the fusion gain is $+0.055 \pm 0.006$ and the trained encoder is ahead of the untrained one by $+0.029 \pm 0.008$, but the untrained encoder itself gains $+0.027 \pm 0.003$ over its six initialisations: with 74,398 training stars, 128 random columns are no longer redundant to a linear fit. Clause (ii) fails, so rotation does not count. On rotation period the gain is $+0.034 \pm 0.002$ on 1,318 test stars, the untrained control is at -0.003 and the paired difference is $+0.037 \pm 0.003$, so it counts. The two sensitivity rows agree with the headline on both verdicts. The dynamics-off contrast is positive on both tasks ($+0.019 \pm 0.007$ and $+0.020 \pm 0.003$), and a paired star-bootstrap of every contrast in the table (1,000 resamples, as in Appendix G) agrees with every seed-band verdict.

Table 6: The two rotation tasks on the rotation pool. The “all stars” rows are those of Table 1; linear and XGBoost readouts and the two supervised baselines are as there. Gain = features $\oplus \mu$ minus features, mean over six encoder seeds, bold beyond 2 SE; untrained gain = the same with the untrained encoder. Sensitivity rows refit the linear readout on the restricted population. n : test stars; prev.: rotator prevalence.

task	population	n (prev.)	linear readout					XGBoost		supervised	
			features	μ	fusion	gain	untr. gain	features	fusion	Conv1D	MLP
rotation (PR-AUC)	all stars	15944 (0.130)	0.386	0.373	0.441	+0.055	+0.029	0.518	0.515	0.409	0.420
	no flare stars	15670 (0.119)	0.356	0.352	0.413	+0.057	+0.029	–	–	–	–
	strict	14182 (0.129)	0.367	0.372	0.427	+0.061	+0.036	–	–	–	–
rotation period (R^2)	all stars	1318	0.718	0.636	0.752	+0.034	−0.003	0.829	0.825	0.697	0.773
	no flare stars	1174	0.713	0.628	0.744	+0.031	−0.006	–	–	–	–
	strict	1145	0.718	0.636	0.748	+0.031	−0.004	–	–	–	–